A Cat's Sense of Smell

Cats have sensitive noses that they use for a variety of purposes, including finding mates and identifying the territory of other cats. Given their reliance on smell, it seems remarkable that the first scientific account of cats using scent in hunting was published in 2010. This study showed that cats do indeed locate prey using the scent marks of prey. Many of the rodents that cats hunt, especially mice, communicate with one another using scent signals carried in their urine. As mammals, cats and mice have noses that work in much the same way, so it is highly unlikely that mice can disguise their scent marks so that cats cannot detect them. Australian biologists proved this by collecting sand from mouse cages and placing it on the ground on the sides of roads. Almost all these patches of sand were visited by predators-mostly foxes, but tracks of wild cats were also apparent-while clean sand was not. The collected data from the experiment did not show how far away the cats had traveled from, but it is possible that significant distances were involved-that is, the cats probably navigated upwind toward the odor sources, rather than only investigating the patches of sand because they looked unusual. We know that many dogs prefer to hunt using their noses and can detect and then locate sources of odor from manv meters awav. While cats prefer to use vision when hunting in daylight, they probably switch to using their sense of smell when hunting at night, when sight, even their sensitive night vision, is likely inadequate for hunting.

Finding prey from the odors it produces can be difficult. Scent marks rarely indicate the current position of the animal that left them, only where it was when it made the mark, perhaps hours earlier. In the sand experiment, the urine samples continued to attract cats for at least a couple of days. It's possible that cats, as sit-and-wait hunters, use the scent marks to see whether they will attract other members of the same species that made them. Mice use urine marks to signal to other mice, and the marks contain a great deal of useful information about the mouse that produced them. Thus, the animal leaving the scent mark doesn't put itself at risk so much as other members of the same species that show up to examine the mark.

Moreover, odors spread out from their sources in ways that are not predictable. We know intuitively that light travels in straight lines but not around obstacles, and that sound travels in all directions, including around obstacles. However, because we rely so little on odor to give us directional information, it's not immediately obvious what problems animals face when determining where an odor is emanating from. Of course, odors outdoors are carried by the air-downwind, not upwind-but air movements close to the ground, where cats operate, are usually highly complex. While the wind may be blowing in a consistent direction a few yards above the ground, friction caused by its contact with the ground and especially with vegetation causes it to break up into eddies (fast-moving circles of air) of various sizes. These carry "pockets" of odor away from the source, so that a cat somewhere downwind of a mouse's nest will get intermittent bursts of mouse smell.

Tracing these bursts to their source, especially in thick vegetation, is likely to require diligent searching and possibly some backtracking. Once a cat has located a source of mouse odor, it is potentially able to use the fact that odors don't travel upwind to position itself downwind of the odor source so its own smell can't be detected by the mouse and then wait for mice to turn up. While sit-and-wait is a well-documented hunting method used by cats, we do not know whether cats routinely prefer patrolling the downwind side of, for example, hedgerows (rows of plants) to avoid their own scent betraying their presence. However, it seems likely that a predator as smart as a cat could quickly learn this tactic, even if it is not instinctive.